

IMPORTS

```
import pandas as pd
import numpy as np
```

SERIES UNIQUE / SORT

```
s.unique()
s.nunique()
s.value_counts()
s.sort_values()
s.sort_index()
s.drop_duplicates()
```

DATETIME COLUMNS

```
scientists['born_dt'] =
    pd.to_datetime(scientists['Born'],
        format='%Y-%m-%d')
scientists['died_dt'] =
    pd.to_datetime(scientists['Died'],
        format='%Y-%m-%d')
scientists['age_days'] = scientists['died_dt'] -
    scientists['born_dt']
scientists['age_years'] =
    scientists['age_days'].dt.days // 365
```

CREATE SERIES

```
pd.Series(['banana', 42])
pd.Series(['Wes McKinney', 'Creator of Pandas'],
    index=['Person', 'Who'])
pd.Series({'goat': 4, 'amoeba': np.nan})
pd.Series(5, index=['a', 'b', 'c'])
```

BOOLEAN SUBSETTING

```
ages > ages.mean()
ages[ages > ages.mean()]
ages[mask]
scientists[scientists['Age'] >
    scientists['Age'].mean()]
scientists[(scientists['Age'] > 40) &
    (scientists['Occupation'] == 'Chemist')]
scientists.query('Age > 40 and Occupation ==
    "Chemist"')
&
|
~
```

CREATE DATAFRAME

```
pd.DataFrame({'Name': [...], 'Age': [...]})
pd.DataFrame(data={...}, index=[...], columns=[...])
pd.DataFrame([['n1', 'n2', 'n3', 'n4']], columns=['A', 'B',
    'C', 'D'])
```

SERIES TRANSFORM

```
s.isin([...])
s.apply(func)
s.map(func_or_dict)
s.astype(dtype)
s.replace(a, b)
s.sample(n)
s.to_frame()
s.corr(other)
s.cov(other)
s.equals(other)
```

DO NOT USE

```
and / or / not on Series
inplace=True
```

SERIES ATTRIBUTES

```
first_row.index
first_row.values
first_row.dtype
first_row.shape
first_row.size
first_row.T
```

VECTORIZED OPERATIONS

```
ages + ages
ages * 2
pd.Series([1,2,3]) + pd.Series([10,20])
ages.values + np.array([...])
```

EXPORT / IMPORT

```
df.to_pickle('scientists_df.pickle')
pd.read_pickle('scientists_df.pickle')
df.to_csv('scientists.csv', index=False)
pd.read_csv('scientists.csv')
df.to_excel('scientists.xlsx', index=False)
pd.read_excel('scientists.xlsx')
df.to_parquet('scientists.parquet', index=False)
pd.read_parquet('scientists.parquet')
```

SERIES METHODS

```
first_row.keys()
s.mean()
s.median()
s.mode()
s.min()
s.max()
s.std()
s.var()
s.quantile(q)
s.describe()
s.sum()
s.count()
```

MODIFY DATAFRAME

```
df['new_col'] = ...
sorted_ages = ages.sort_values()
scientists['Age'] =
    scientists['Age'].sample(frac=1).reset_index(drop=True)
df.rename(columns={'old_name': 'new_name'})
df.drop(columns=['col_to_remove'])
df[['col_b', 'col_a']]
df['col'].apply(lambda x: x * 2)
df.apply(lambda row: row['a'] + row['b'], axis=1)
```